

RAJALAKSHMI ENGINEERING COLLEGE

Experiment 5: EDA - Data Visualization with Matplotlib

Name	Kabilan S	Register No.	241501502
Department	AIML	Year / Semester	III / V

AIM

To explore and understand patterns, distributions and relationships in a dataset through visual representations that support insight discovery and anomaly detection.

SOURCE

EXPT5/exp 5 fdva.pdf

PYTHON CODE

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

df = pd.read_csv("data.csv")
df.head()

x = np.linspace(0, 10, 100)
y = np.sin(x)

plt.plot(x, y, color='blue', label='Sine Wave')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line Plot')
plt.legend()
plt.show()

categories = ['A', 'B', 'C', 'D']
values = [3, 7, 5, 4]

plt.bar(categories, values, color='orange')
plt.xlabel('Category')
plt.ylabel('Values')
plt.title('Bar Plot')
plt.show()

data = np.random.randn(1000)

plt.hist(data, bins=20, color='purple', edgecolor='black')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Histogram')
plt.show()
```

EXECUTED OUTPUTS

The original executed table and Matplotlib line, bar and histogram outputs are preserved on the following pages.

RESULT

The given visualization program was written and executed successfully.

V. EDA – DATA VISUALIZATION

AIM:

To explore and understand the underlying patterns, distributions, and relationships within the dataset through visual representations, which aids in uncovering insights, detecting anomalies, and guiding further analysis.

```
[1]: import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
```

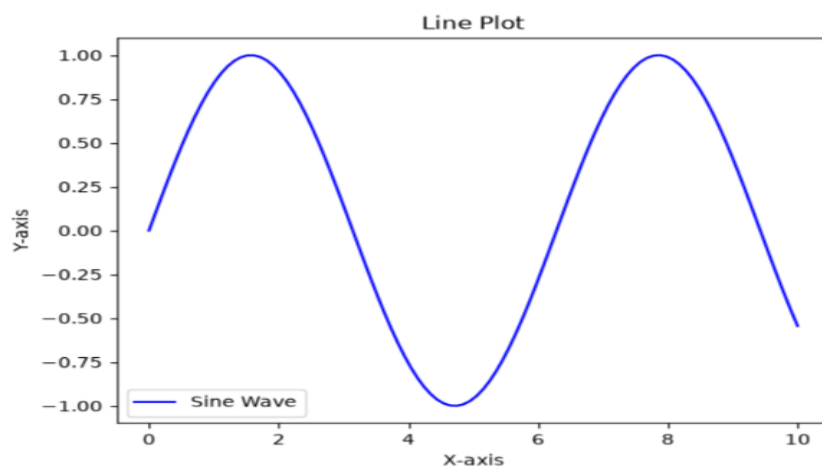
```
[2]: df = pd.read_csv("data.csv")
df.head()
```

```
[2]:
```

	Name	Age	Marks	Department
0	Arun	19	85	CSE
1	Priya	20	92	AIML
2	Kavin	19	78	ECE
3	Divya	21	88	CSE
4	Rahul	20	75	AIML

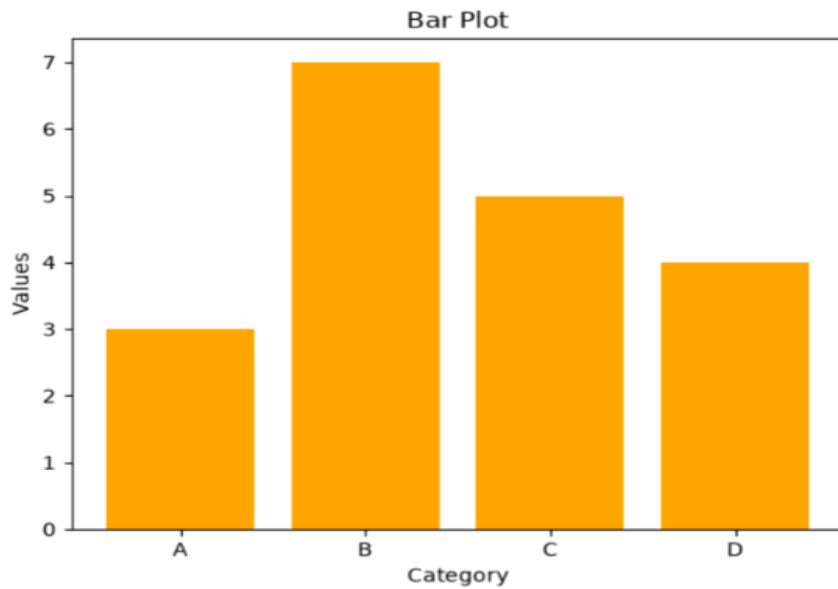
```
[3]: x = np.linspace(0, 10, 100)
y = np.sin(x)

plt.plot(x, y, color='blue', label='Sine Wave')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line Plot')
plt.legend()
plt.show()
```



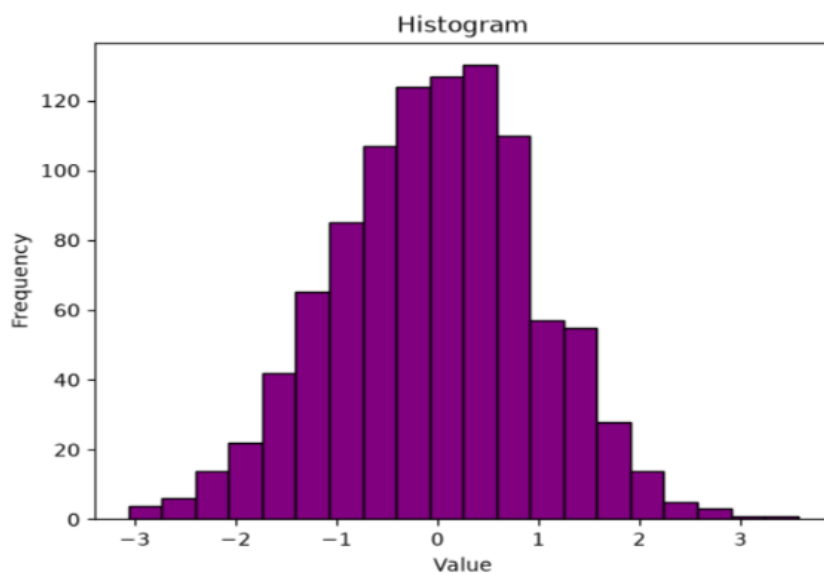
```
[4]: categories = ['A', 'B', 'C', 'D']
values = [3, 7, 5, 4]

plt.bar(categories, values, color='orange')
plt.xlabel('Category')
plt.ylabel('Values')
plt.title('Bar Plot')
plt.show()
```



```
[5]: data = np.random.randn(1000)

plt.hist(data, bins=20, color='purple', edgecolor='black')
plt.xlabel('Value')
plt.ylabel('Frequency')
plt.title('Histogram')
plt.show()
```



RESULT:

Thus, the given program was written and executed successfully.